CO5 - AT3

Conversion of the flow chart to the c-coding.

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course:- CSA0315
Data structures.

1) convert a flowchart for Depth First search using stack into c-coding.

2) A flowchart implements shortest path algorithm and Dijkstra's algorithm convert it into c-programming.

3) A flowchart performs Prim's Algorithm convert flowchart into recursive c-programming

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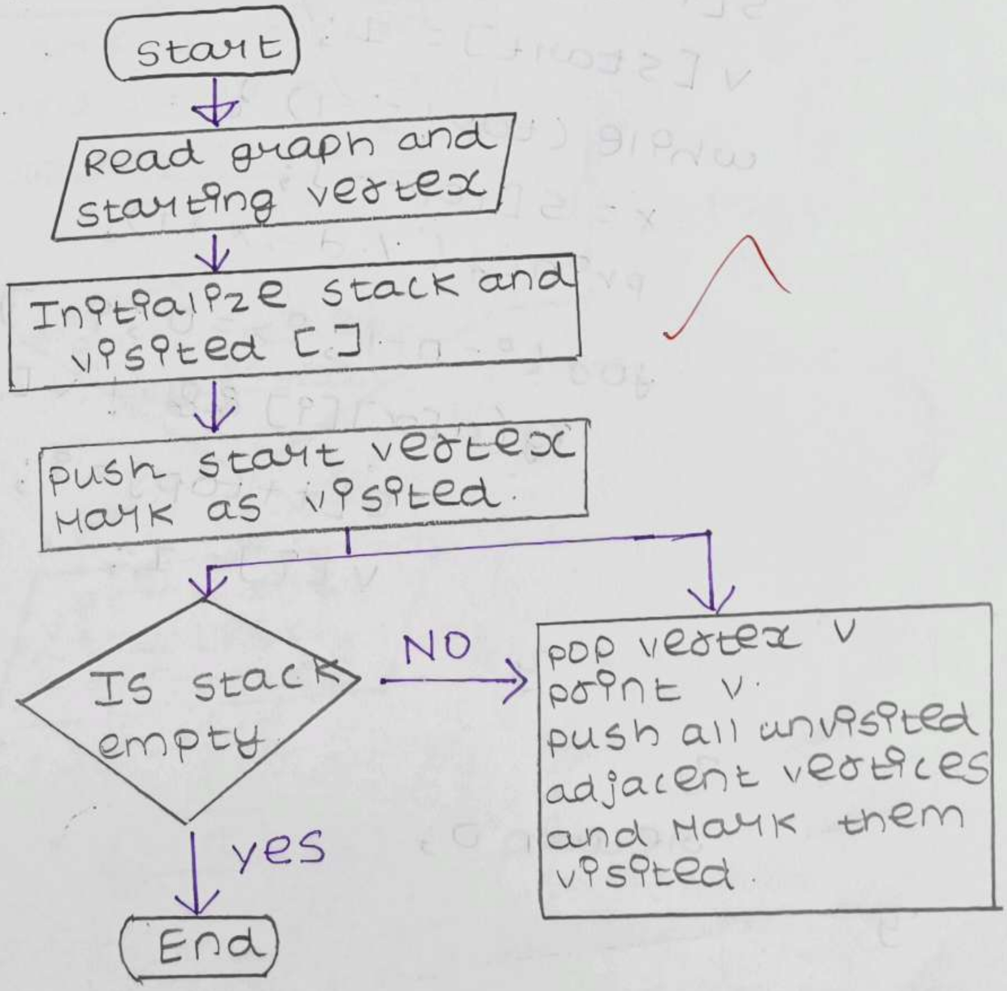
1) DFS using stack.



C-code:

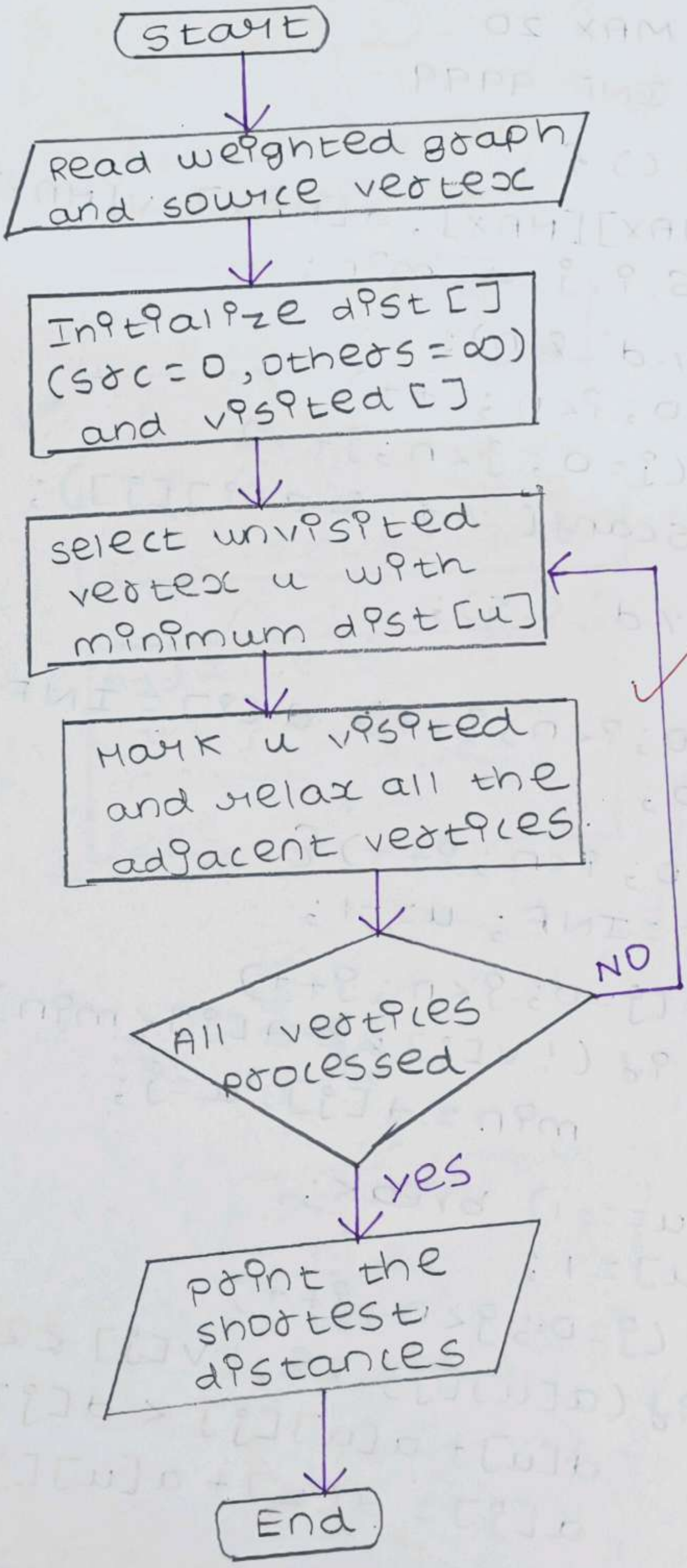
```
#include <stdio.h>
#define MAX 20
int main() {
int a[MAX][MAX], v[MAX] = {0}, s[MAX];
  int n, start, top = -1, i, x;
  scanf ("%d", &n);
  for (i=0; i<n; i++)
      for (j=0; j<n; j++)
          scanf("%d", & a[i][j]);

  scanf("%d", & start);
  start--;
  s[++top] = start;
  v[start] = 1;
  while (top != -1) {
     x = s[top--];
     printf ("%d", x+1);
     for (i=n-1; i>=0; i--)
        if(a[x][i] && ! v[i]) {
              s[++top] = i;
              v[i] = 1;

        }

  }
  return 0;
}
```

Dijkstra's shortest path:



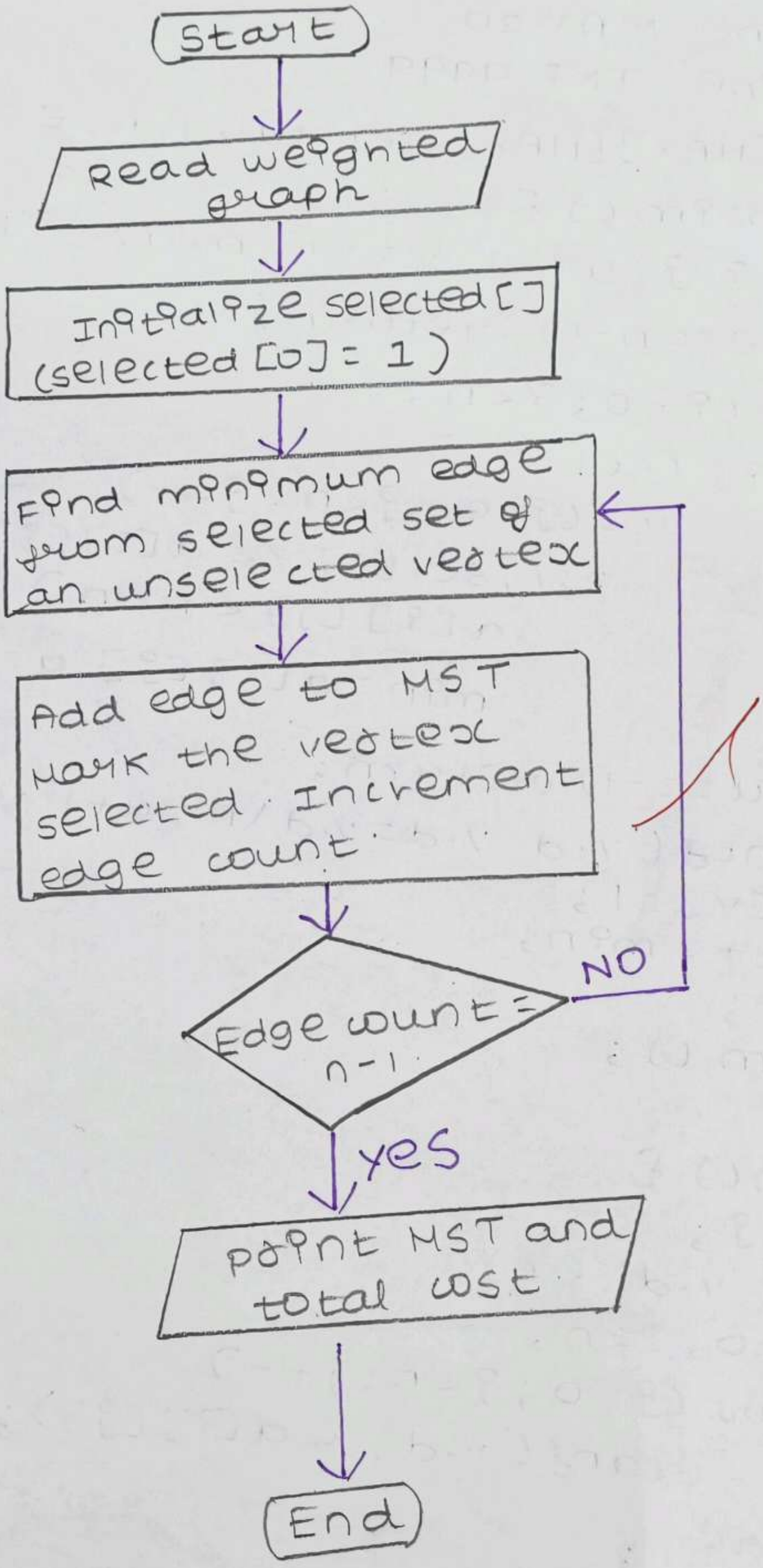
c-code:-

```c
#include <stdio.h>
# define MAX 20
# define INF 9999
int main () {
  int a[MAX][MAX], d[MAX], v[MAX] = {0};
  int n, s, i, j, u, min;
  scanf("%d", &n);
  for (i=0; i<n; i++)
      for (j=0; j<n; j++)
          scanf("%d", & a[i][j]);

  scanf("%d", & s);
  s--;
  for (i=0; i<n; i++) d[i] = INF;
  d[s]=0;
  for (i=0; i<n; i++) {
      min=INF; u=-1;
      for(j=0; j<n; j++)
          if (!v[j] && d[j]<min)
              min = d[j], u=j;

      if (u==-1) break;
      v[u]=1;
      for (j=0; j<n; j++)
          if (a[u][j] && !v[j] &&
              d[u]+a[u][j] < d[j])
              d[j] = d[u] + a[u][j];
  }
  for (i=0; i<n; i++)
      printf("%d -> %d/n", i+1, d[i]);
  return 0;
}
```

Prim's Algorithm:-



C-code:-

```c
#include <stdio.h>
#define MAX 20
#define INF 9999
int a[MAX][MAX], sel[MAX], n, e=0, cost=0;
void prim() {
    int i, j, u=-1, v=-1, min=INF;
    if (e==n-1) return;
    for (i=0; i<n; i++)
        if (sel[i])
            for (j=0; j<n; j++)
                if (!sel[j] && a[i][j] &&
                    a[i][j] < min)
                    min=a[i][j], u=i, v=j;

    if (u==-1) return;
    printf("%d-%d=%d\n", u+1, v+1, min);
    sel[v]=1;
    cost+=min;
    e++;
    prim();
}
int main() {
    int i,j;
    scanf("%d", &n);
    for (i=0; i<n; i++)
        for (j=0; j<n; j++)
            scanf("%d", &a[i][j]);

    sel[0]=1;
    prim();
    printf("cost = %d", cost);
    return 0;
}
```